TECHNOLOGY CAPSTONE PROJECT | MIS 4500

Personal Health Record Management and Sharing System



Project Proposal Presentation

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Presentation Outline

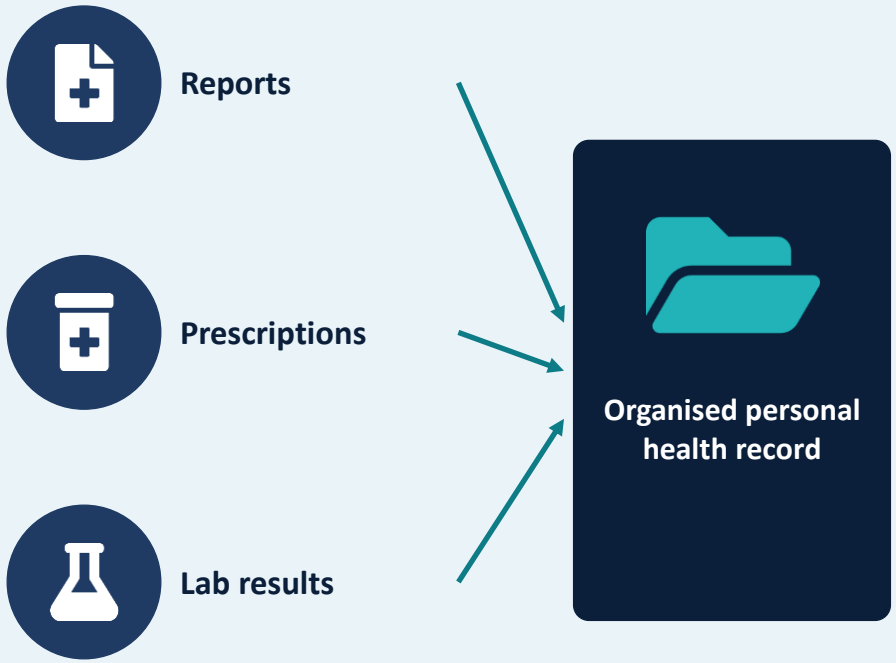
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Following the lecturer's required order

Introduction

- Patients may receive reports, prescriptions and other medical documents from different healthcare providers.
- Keeping these documents organised and accessible can be difficult.
- Relevant information may need to be retrieved during medical consultations.
- This project proposes a web-based system to organise health information, review submitted documents administratively and share authorised records with doctors.



Documents from different sources, brought into one organised record

Problem in Brief



Scattered documents

Medical documents can be scattered across different sources.



Hard to maintain history

Patients may find it difficult to maintain an organised health history.



Inconvenient retrieval

Retrieving relevant records for consultations can be inconvenient.



Several users, clear rules

Multiple users need clearly defined permissions and a structured document-handling process.

CENTRAL PROBLEM

How can a locally runnable web application integrate medical document submission, administrative review, chronological record management and patient-authorised doctor access?

Literature Review



Roehrs et al. (2017)

Personal health records: A systematic literature review

What it covers

The review examines PHR functions, data sources, architecture and implementation challenges.

Relevance to this project

Supports bringing personal health information from several sources into one organised, accessible record, the focus of this project.



Jayabalan and O’Daniel (2016)

Access control and privilege management in electronic health records

What it covers

The review covers role-based and context-specific access models and the need for audit trails.

Relevance to this project

Supports separate permissions for patients, doctors and the review team, patient-controlled sharing and basic access logging.

Existing Systems and Research Gap

Feature	HealthSync	NexusCare	Medica	Proposed system
Health record management	**Available**	**Available**	**PHR described**	**Planned**
Medical document submission	**Available**	Lab upload described	Not confirmed	**Planned**
Record sharing / access management	**Available**	**Role-based access described**	Not confirmed	**Patient-authorised access**
Administrative review workflow	Not confirmed	Not confirmed	Not confirmed	**Planned**
Chronological health history	**Timeline view described**	Not confirmed	Not confirmed	**Planned**
Consultation-ready summary	Not confirmed	Not confirmed	Not confirmed	**Planned**

RESEARCH GAP AND CONTRIBUTION

Personal health record storage and sharing already exist. This project focuses on implementing and evaluating an integrated workflow: document submission, administrative review, chronological record management and patient-authorised doctor access.

"Not confirmed" = not confirmed in reviewed sources; it does not mean the platform lacks the feature. Based on public pages reviewed in October 2026.
Sources: HealthSync help pages on uploading documents and sharing health records (HealthSync, n.d.-a, n.d.-b); NexusCare (n.d.); Medica (n.d.).

Aim



“To design, develop and evaluate a web-based Personal Health Record Management and Sharing System that enables patients to organise their medical information, manage document reviews and share authorised records with doctors.”

Project aim from the approved proposal

Objectives

1. Analyse the requirements of patients, doctors and the designated document review team.
2. Design a user-friendly web application with separate interfaces and role-based permissions.
3. Implement medical document submission, administrative review and approval workflows.
4. Develop chronological patient health records and patient-controlled doctor access.
5. Develop consultation-ready summaries from selected patient records.
6. Test functionality, access permissions and usability using fictitious data.

Proposed Solution



Patient

- Submit medical documents
- Enter health information manually
- View submission statuses
- View the chronological health record
- Manage authorised doctor access



Document Review Team

- Review submitted documents
- Approve or reject submissions
- Request corrections
- Record review outcomes



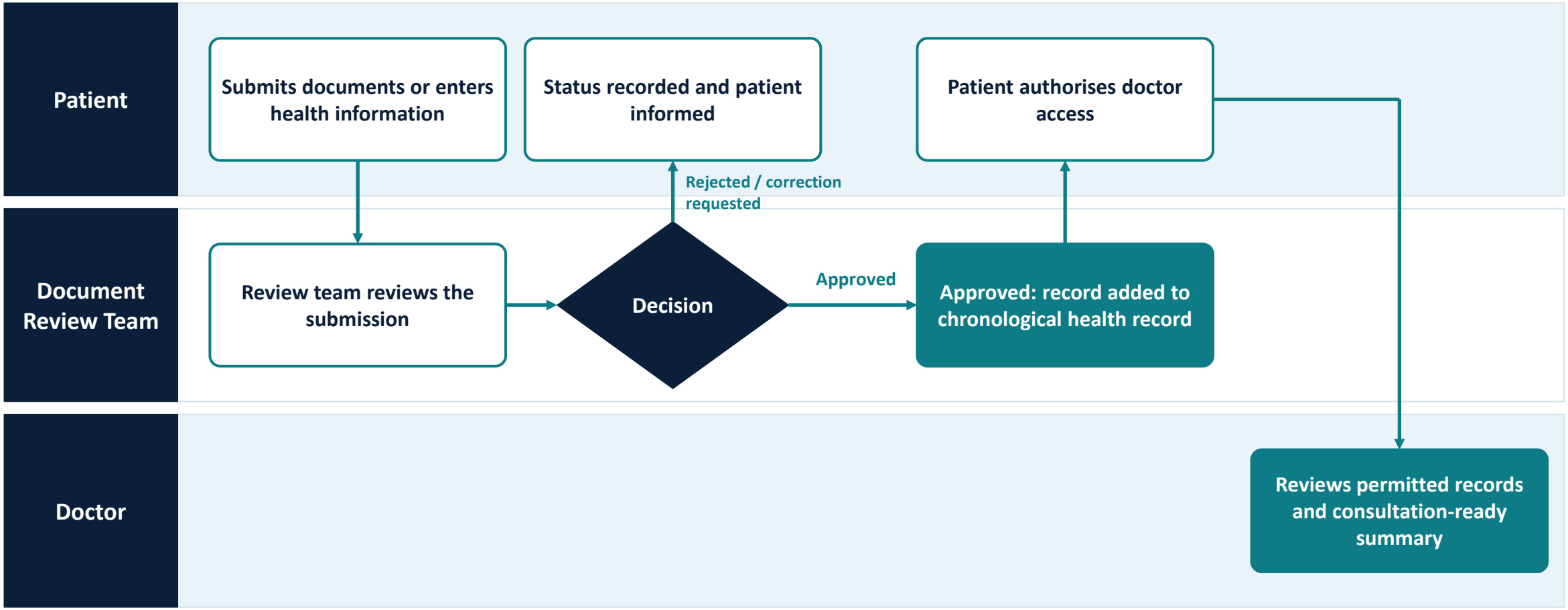
Doctor

- Access patient-authorised records
- Review relevant health history before consultation
- Use a summary generated from selected records

Approved records are added to the patient's organised health history. Rejected submissions and correction requests follow a separate path.

Administrative approval does not mean clinical verification.

Proposed System Workflow



Administrative approval is not clinical verification. The workflow involves no automatic medical verification and no hospital integration.

Technologies and Resources

Potential technologies

Frontend	HTML, CSS, JavaScript and potentially Bootstrap
Backend	Python with Flask as a potential option
Database	MySQL or another suitable database
Code editor	Visual Studio Code or an equivalent IDE
Diagramming	Draw.io
Testing	Web browser and suitable development / testing tools
Documentation	Microsoft Word



Hardware

- Laptop or desktop computer
- Sufficient memory and storage for development and testing
- Internet connection for research and obtaining development tools

The final technology stack is not yet fixed and may change based on compatibility, technical requirements and development progress.

The prototype will run locally. Cloud storage, hosting and online deployment are not part of this project.

Research Methodology

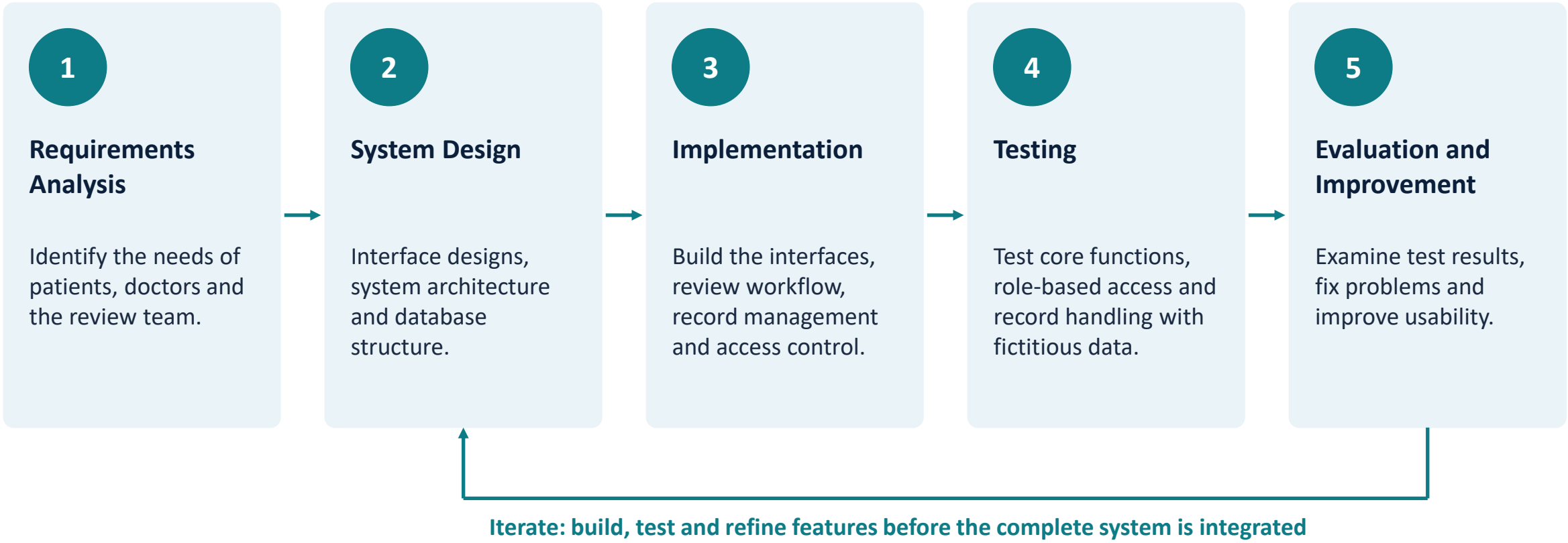
Applied research approach

- Review academic literature on personal health records and access control.
- Examine existing digital health platforms.
- Identify requirements for patients, doctors and the review team.
- Use the findings to guide system design and requirements.
- Carry out functional testing and, where feasible, usability evaluation using fictitious data.



Evidence base so far: literature and existing-platform review. No interviews, surveys or user studies have been conducted.

Development Methodology



Proposed approach. The prototype is still to be developed and will be demonstrated in a local environment.

Future Work

Within this capstone

- Locally runnable academic prototype
- Fictitious patient data only
- Patient, doctor and review-team roles
- Document review, chronological records and patient-authorised access
- Consultation-ready summaries
- Functional and usability testing

Potential future directions

- Improved document search and filtering
- Additional personal health metrics
- Enhanced consultation-summary presentation
- Further usability improvements
- Possible integration with external healthcare systems, subject to feasibility, permissions, privacy and security requirements

These are possible future directions only. They are not commitments for the current capstone project.

Project Plan (Proposed Schedule)

Phase	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Requirements analysis														
Literature review and system research														
System design and database planning														
Prototype: user interface development														
Prototype: core functionality development														
System integration														
Functional testing														
Usability / user evaluation														
Refinements														
Final report and documentation														
Final demonstration														



Proposed schedule in relative weeks. Course dates were not supplied, so none are assumed, and no activity is marked complete.

Conclusion

- Addresses the organisation and controlled sharing of personal health information.
- Combines document submission, administrative review, chronological records and patient-authorised doctor access.
- Will be developed as a locally runnable functional prototype.
- Will use fictitious data only.
- Functionality and usability will be evaluated within the defined academic scope.

EXPECTED OUTPUT



A working prototype demonstrating the proposed integrated workflow.

References

HealthSync. (n.d.-a). *Uploading documents*. https://www.healthsync.lk/help/uploading-documents/

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Medica. (n.d.). *Medica: Smart healthcare platform*. https://medica.lk/

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Roehrs, A., da Costa, C. A., da Rosa Righi, R., & de Oliveira, K. S. F. (2017). Personal health records: A systematic literature review. *Journal of Medical Internet Research, 19*(1), Article e13. https://doi.org/10.2196/jmir.5876

Thank You

Questions?